Vincent Costa

vincentcosta@college.harvard.edu • personal website • https://github.com/vincentcosta04

Education

Harvard University — Cambridge, MA
A.B. in Mathematics. — May 2027

Research Experience

Harvard Math Summer Research Experience for Undergraduates (REU) — Cambridge, MA
Advisor: Professor Chuck Doran — June – August 2026
Project: *Classification of Adinkras via Group Actions, Orbifolds, and Gerbes.* Draft of research manuscript.

Ni Lab, Department of Physics, Harvard University — Cambridge, MA
Advisor: Professor David Wellnitz — June 2025 – Present
Project: *Optimizing Multi-Qubit Gates for Dual-Species Rydberg Arrays*. Preprint in preparation.

Ni Lab, Department of Physics, Harvard University — Cambridge, MA
Advisors: Kenneth Wang, Ryan Cimmino, and Yu Wang — June – August 2024
Projects:

1. Flask web application to interface with a spatial light modulator (SLM). GitHub repository.
2. Python script to apply optimal Zernike polynomials to an SLM to correct optical aberrations. GitHub repository.

Supervised Reading and Projects

Senior Thesis — Cambridge, MA
Advisor: Lorenzo Riva — April 2026 – Present
Tentative title: *What are ∞-categories and what they used for?*

Supervised Reading Course on Quasicategories — Cambridge, MA
Advisor: Professor Mike Hopkins — September 2026 – Present

Appendix for Math 231br Lecture Notes (Topology of Algebraic Manifolds) — Cambridge, MA
Instructor: Thomas Brazelton — March 2026 – Present
Title: *Stacks on Spaces*. Most recent draft.

Harvard Math Directed Reading Program (DRP) — Cambridge, MA
Mentor: Matthew Niemiro — January 2025 – Present

Talks and Presentations

Adinkras, Codes, and Cubes (Notes) — Cambridge, MA
Harvard Math Table (undergraduate seminar) — October 2026

Classification of Adinkras via Action Groupoids — Cambridge, MA
Harvard Math Summer REU Symposium — August 2026

The Eckmann-Hilton Argument (Notes) — Cambridge, MA
Harvard Math Table — March 2026

What Do Finite Pointed Sets Know? — Cambridge, MA
Harvard Math DRP Symposium — May 2025

Teaching and Leadership Experience

Course Assistant , Cambridge, MA

Hold office hours and grade homework for:

- Math 25a: Theoretical Linear Algebra and Real Analysis I — September 2026 – Present
- Math 122: Algebra I: Theory of Groups and Vector Spaces — January – May 2026
- Math 112: Introductory Real Analysis — September – December 2025
- Physics 15a: Introductory Mechanics and Relativity — September 2024 – May 2025

Technical Skills and Languages

Scientific Computing: Julia, Python, NumPy, SciPy, Pandas, Scikit-learn.
Web Development: Flask, JavaScript, HTML, CSS, Bootstrap.
Languages: Native proficiency in English and French.